Applied Machine Learning

Boosting

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CHALMERS

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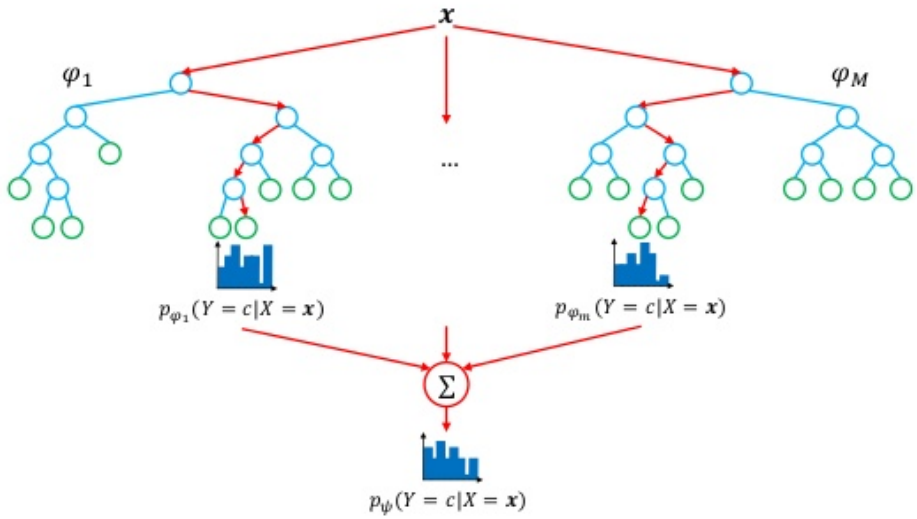
ensembles: recap

- we have seen that **ensembles** can be more robust than single models (particularly for trees)



- we saw some **ensemble building** methods such as bagging
- today: an approach that builds ensembles **iteratively**

can we try to fix the errors?

- assume that we have trained a classifier or regressor, and that it makes some mistakes on the training data
- can we try to train another model that "cleans up the mess"?

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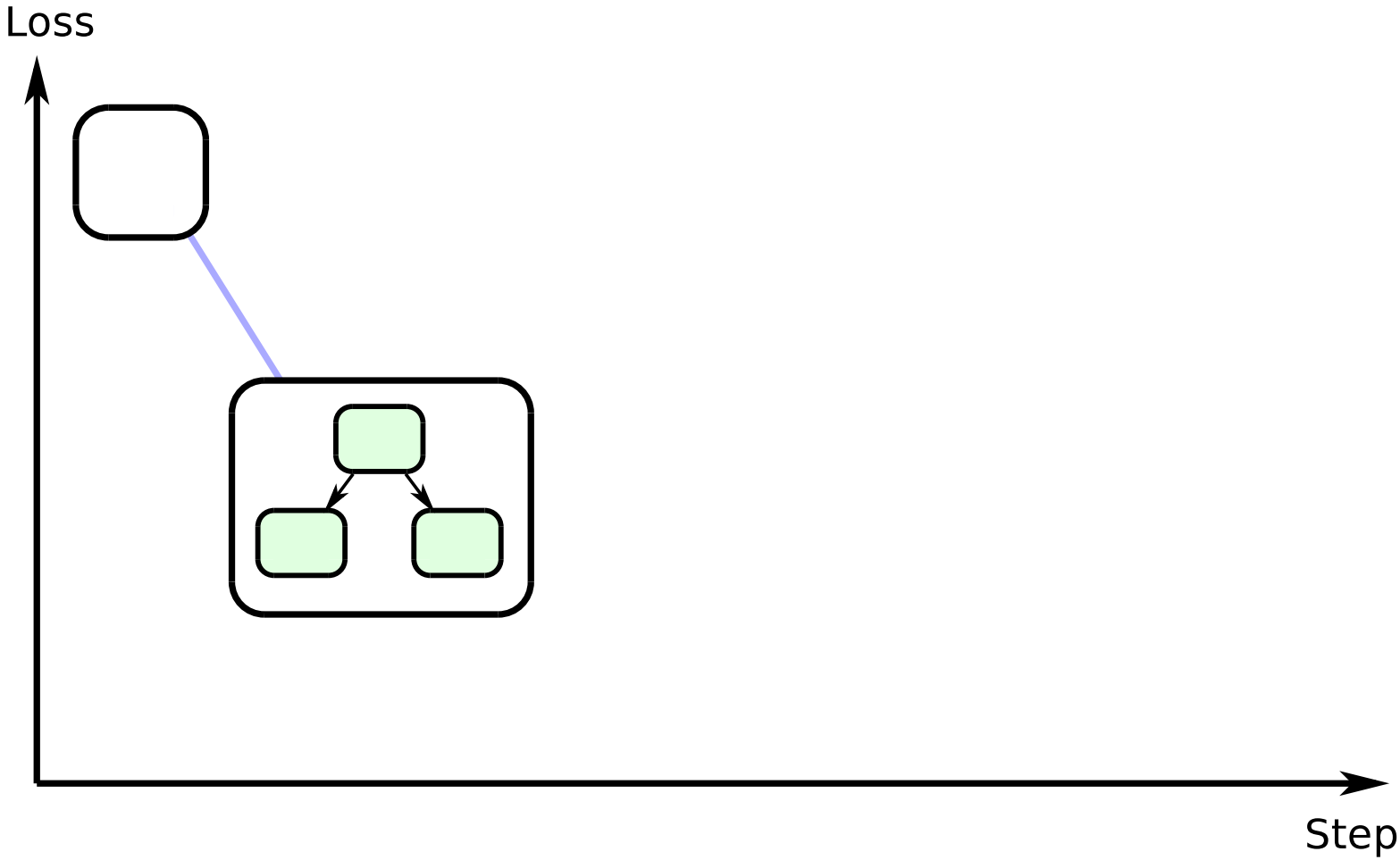
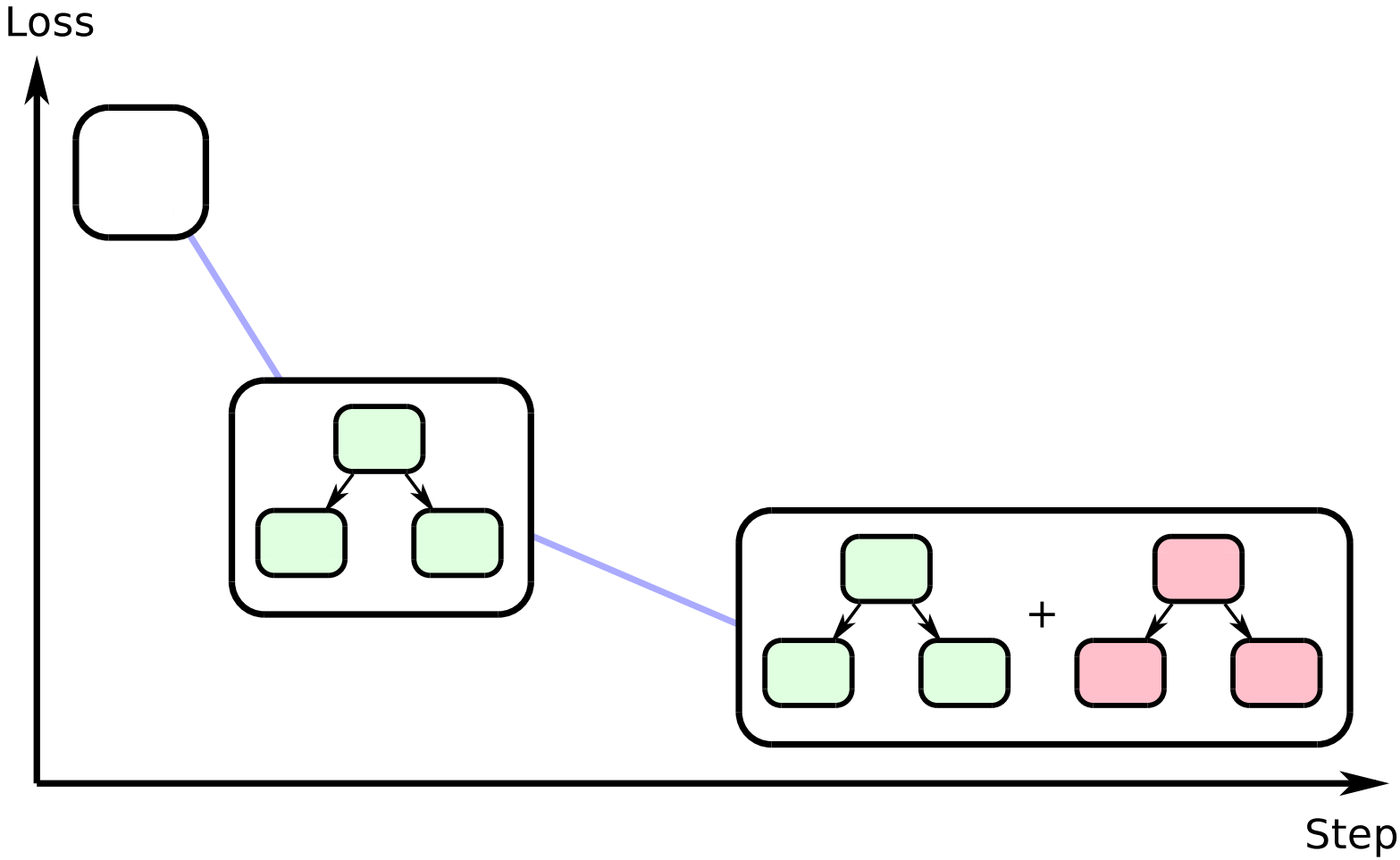
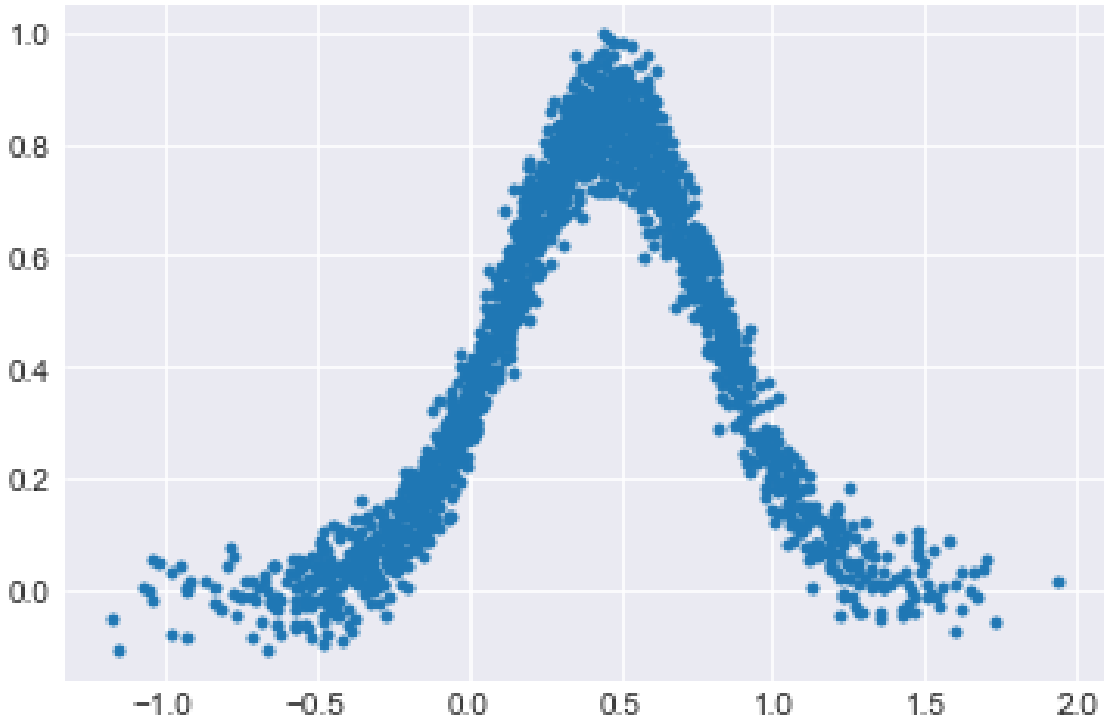
- assume that we have trained a classifier or regressor, and that it makes some mistakes on the training data
- can we try to train another model that "cleans up the mess"?
- yes, and this idea is called **boosting**
  - an ensemble where the sub-models are trained sequentially
  - to gradually improve the overall performance


Loss
Step

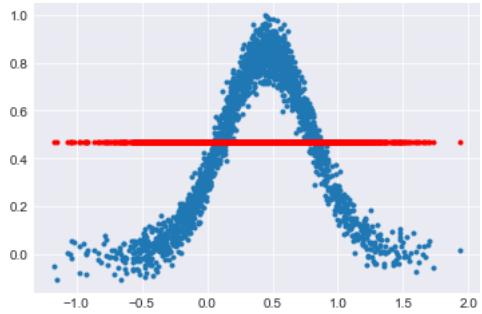

Loss
Step


Loss
+
Step

toy regression example



fixing the errors iteratively



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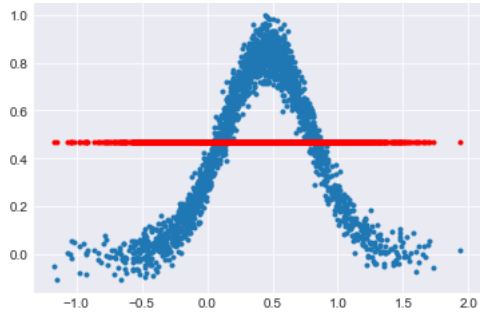

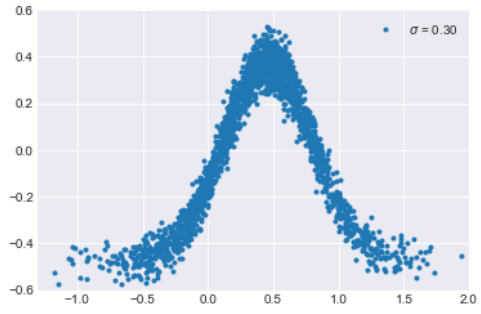

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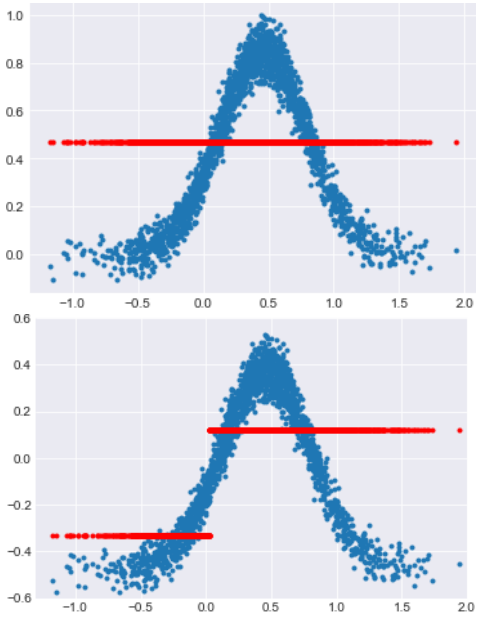

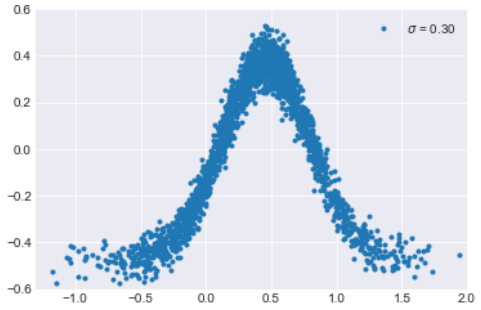

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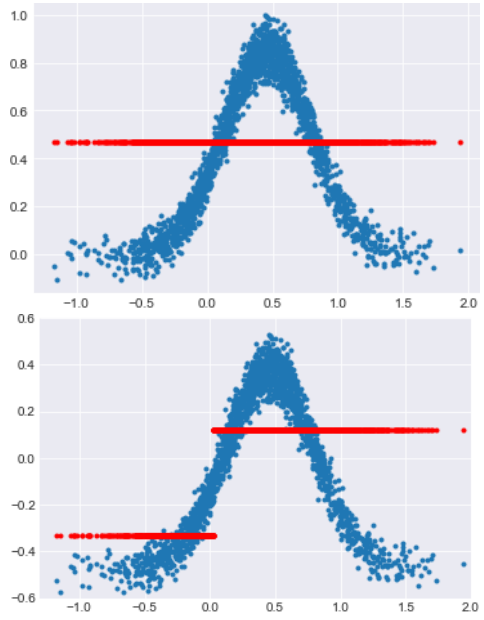
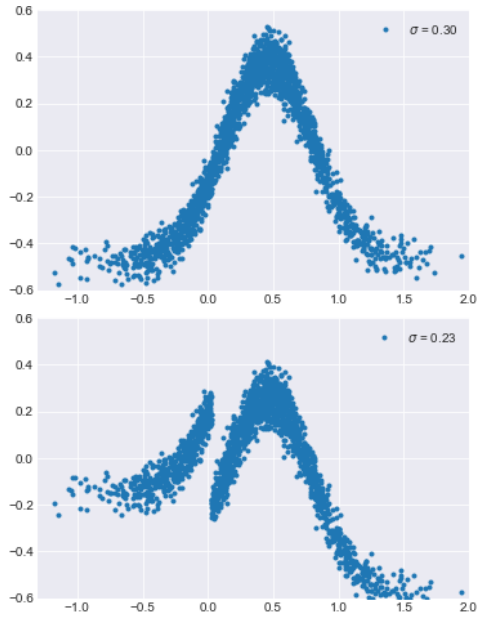
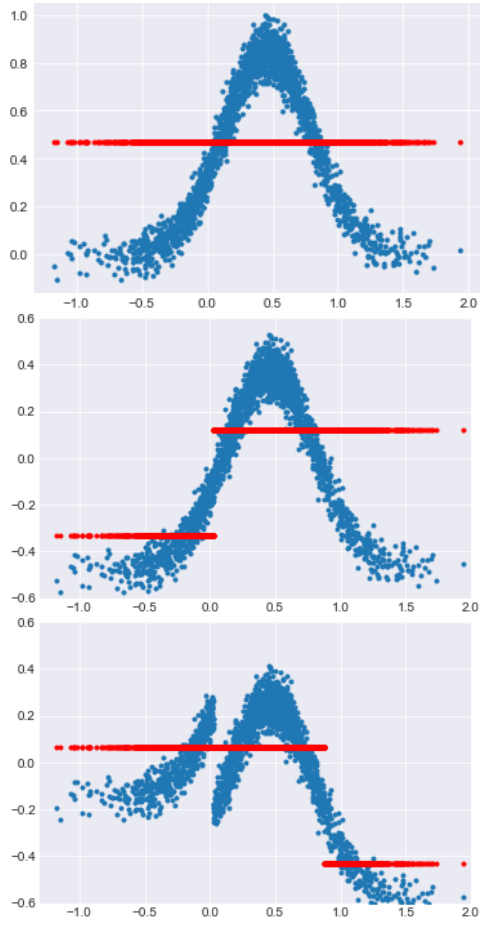
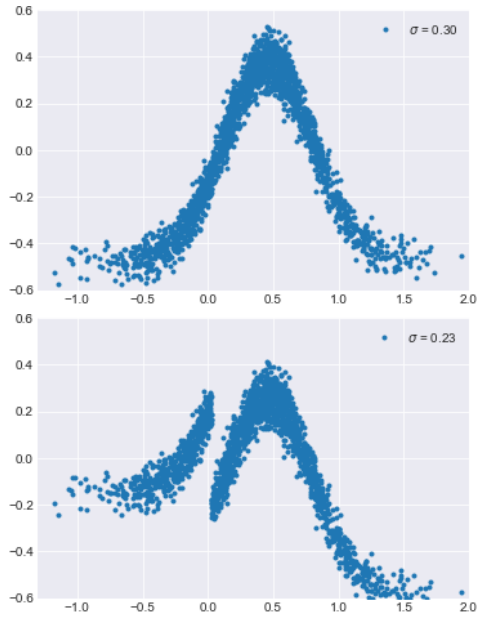




fixing the errors iteratively





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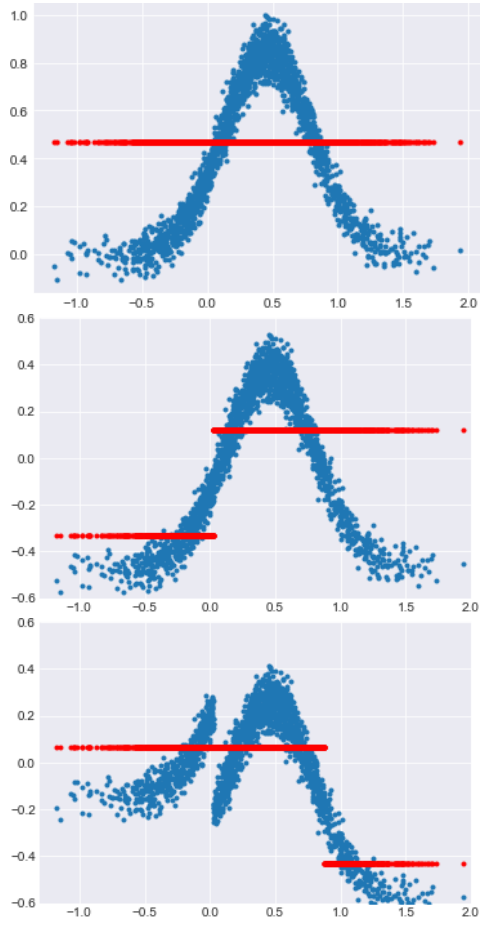

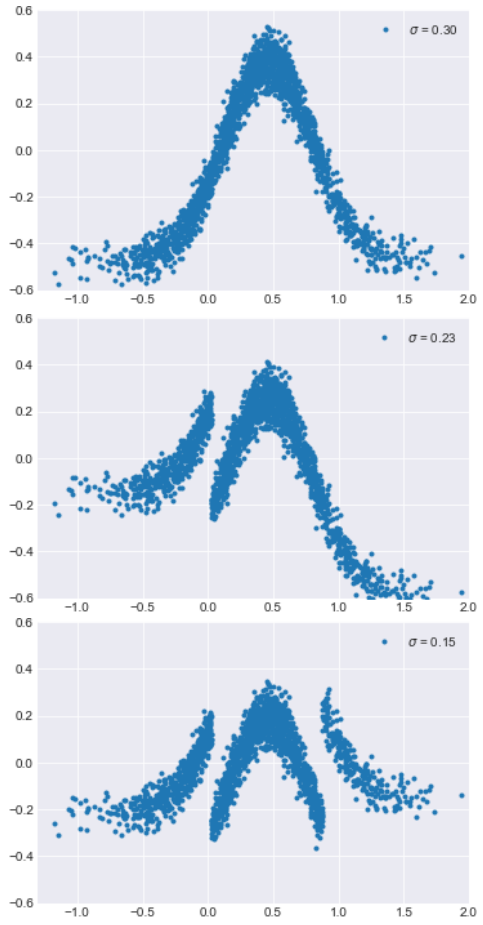

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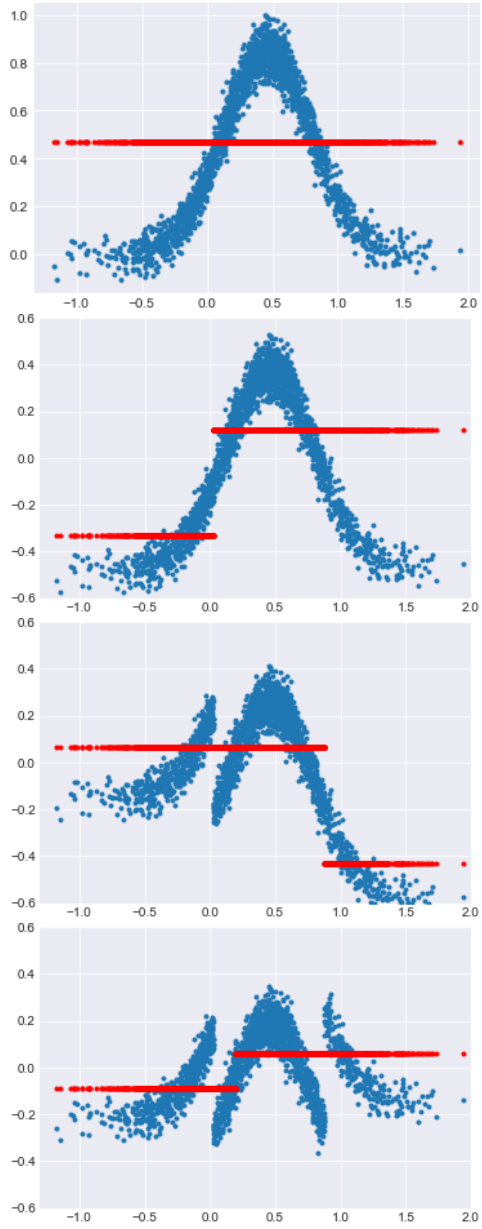
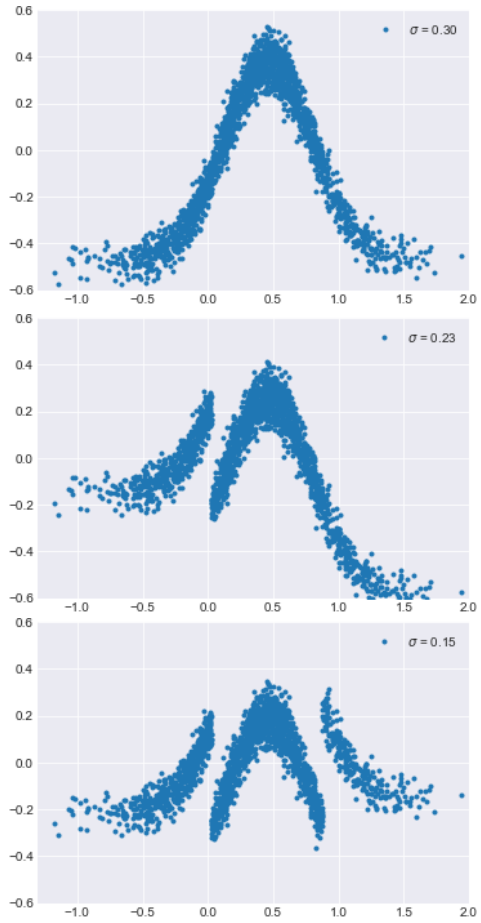
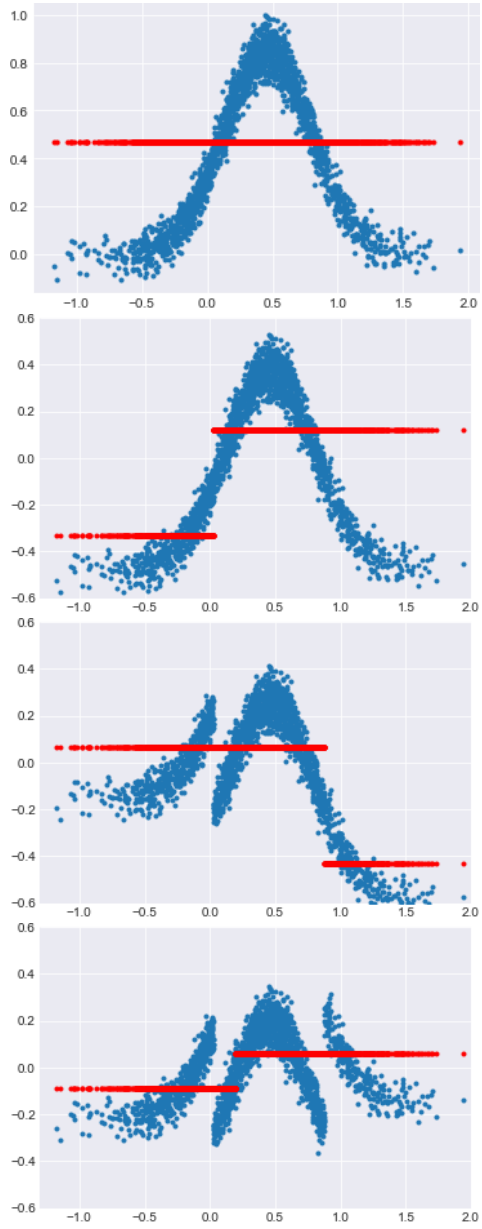
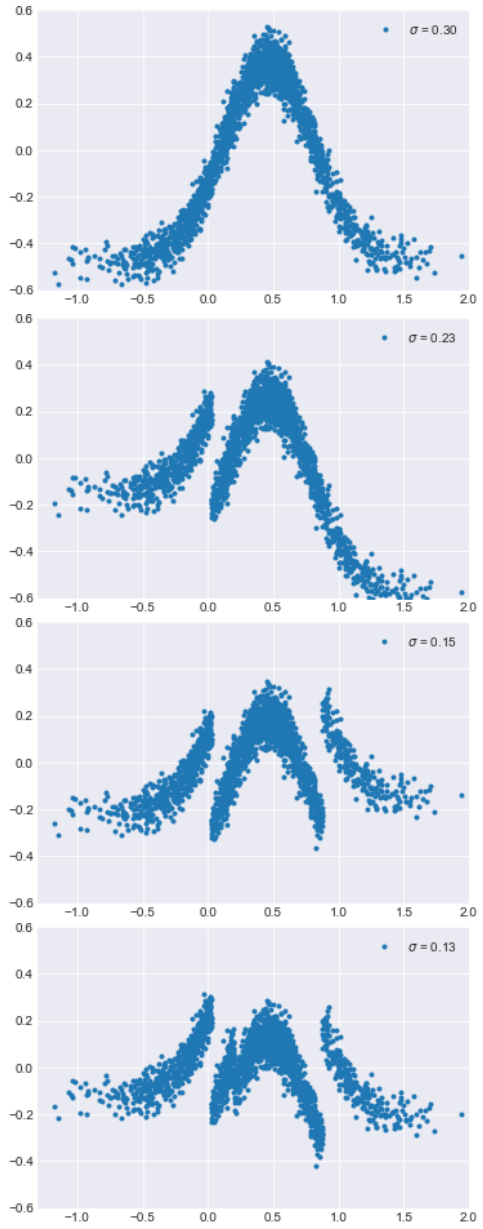


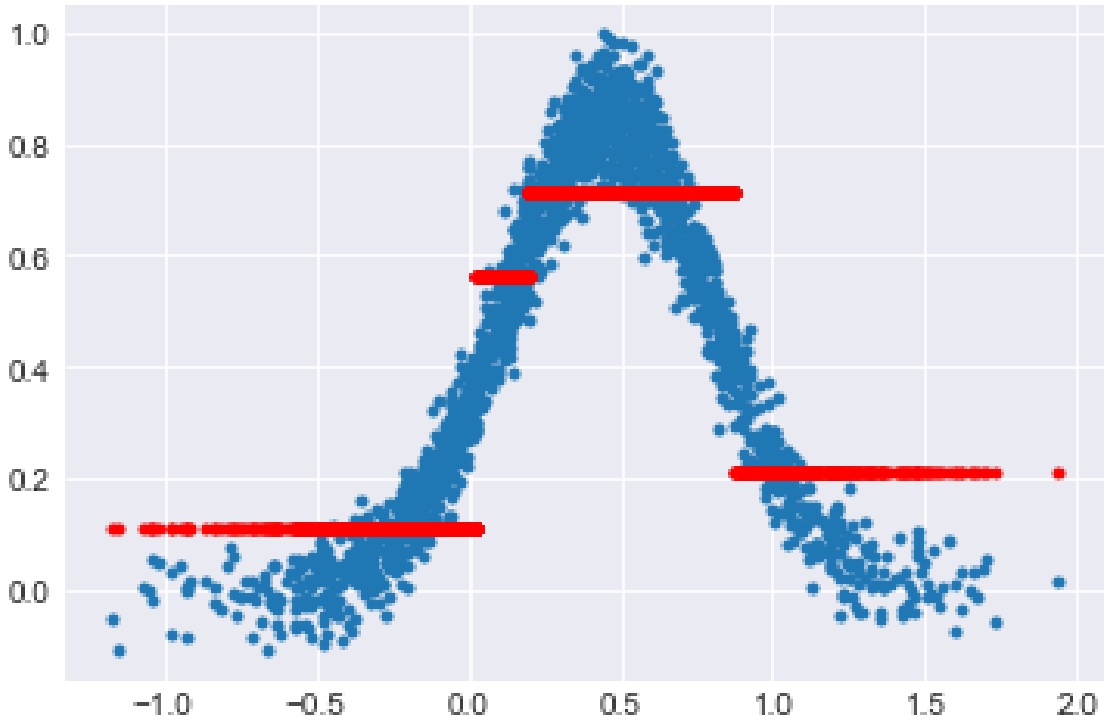


fixing the errors iteratively



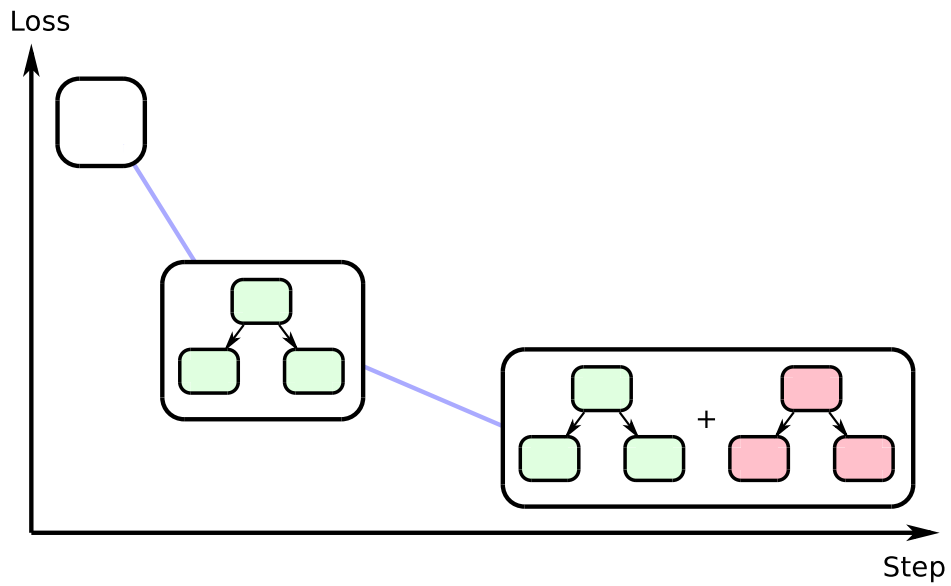


toy regression example: combining the pieces



approaches to boosting: overview

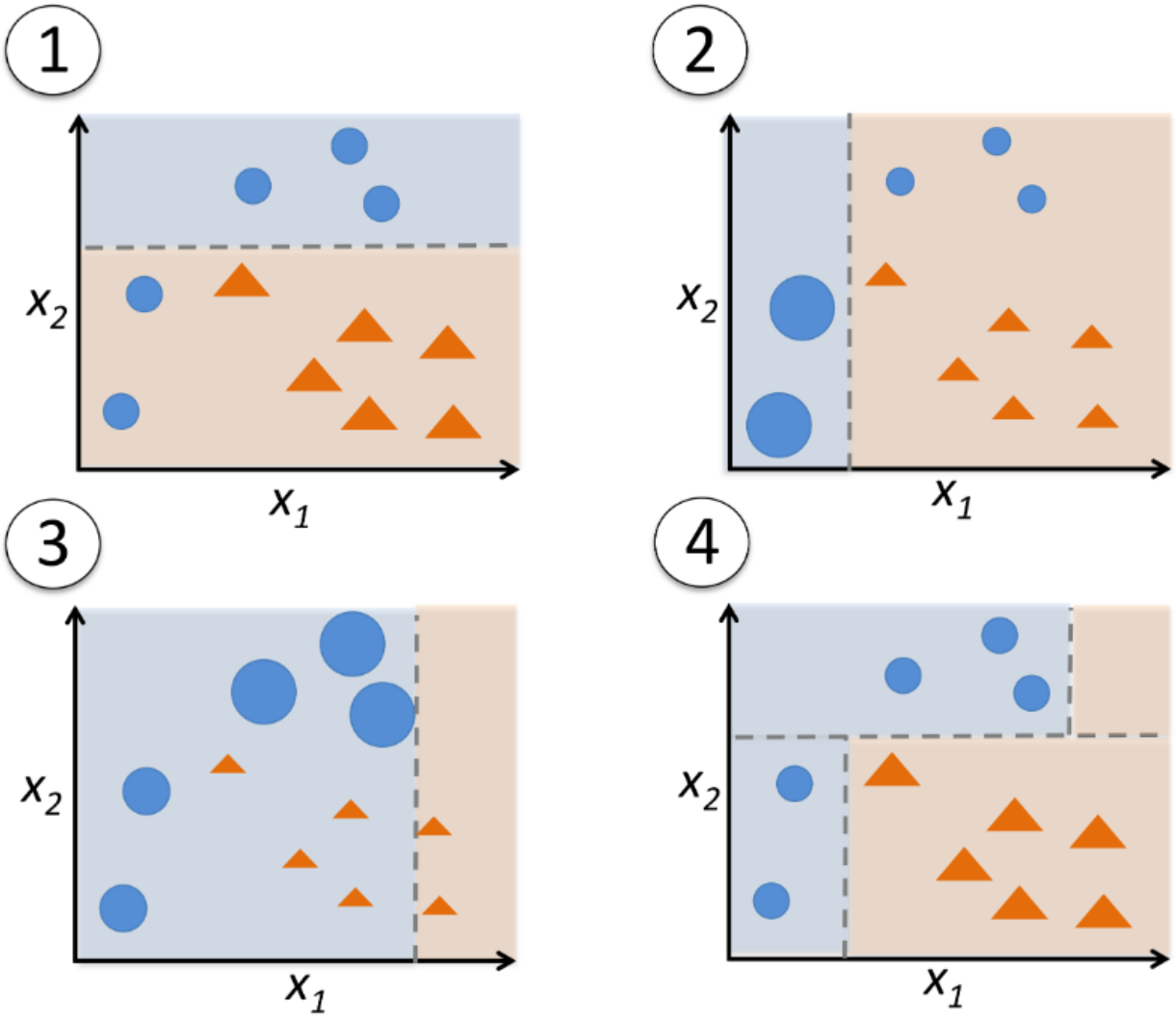
- **AdaBoost**: classical algorithm, works for classification tasks
- **gradient boosting**: generalizes AdaBoost, works for classification and regression



adaptive boosting: AdaBoost

- **AdaBoost** is the most famous boosting algorithm
- after each iteration, check which instances are misclassified
  - then give misclassified instances a higher importance before the next iteration
- this can be applied using any base classifier that can handle weighted instances
  - typical choice: small decision trees ("decision stumps")

adaptive boosting: AdaBoost



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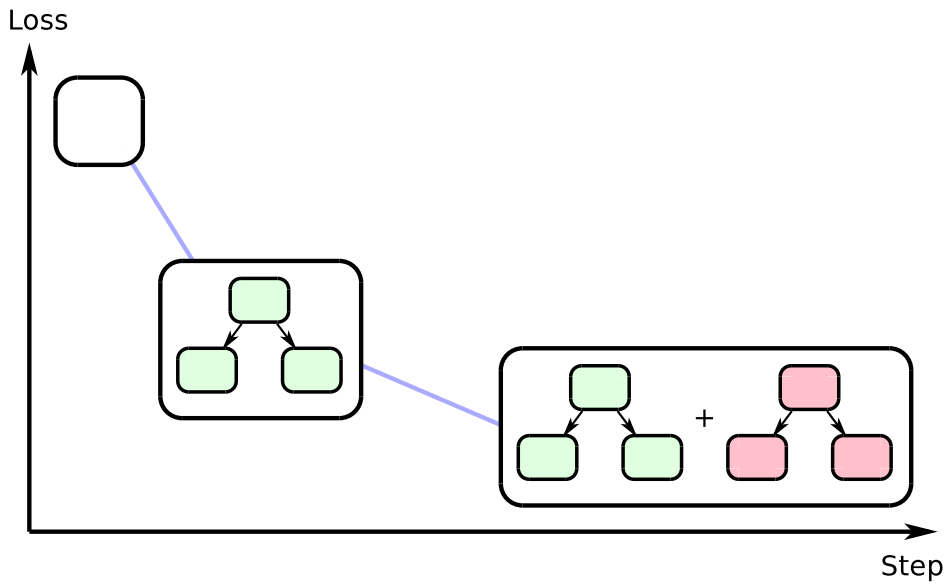
gradient boosting

- **gradient boosting** generalizes AdaBoost
- idea: gradually add sub-models to minimize a loss function
- it is based on a principle of gradient descent



gradient boosting: pseudocode

let F_0 be a "dummy" constant model
for $m = 1, \ldots, M$
  for each pair $(\boldsymbol{x}_i, y_i)$ in the training set
    let r_i be the **pseudo-residual** $R(y_i, F_{m-1}(\boldsymbol{x}_i))$
  train a sub-model h_m on the pseudo-residuals
  create F_m by **adding** h_m to F_{m-1}
return F_M

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- in practice, the sub-models are typically trees
- the **pseudo-residual** $R(y_i, F_{m-1}(\boldsymbol{x}_i))$ is defined as the negative gradient of the loss function

$$-\nabla_{\hat{y}} \text{Loss}(y, \hat{y})$$

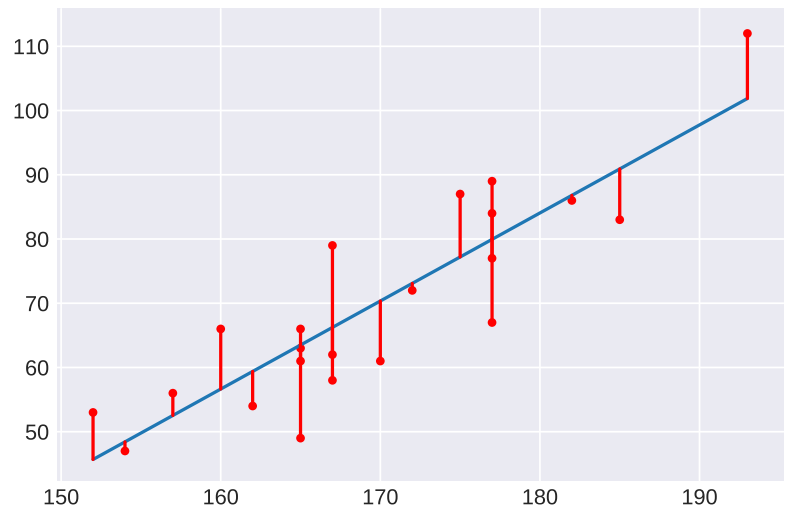
why "pseudo-residual"?

- for the squared error loss

$$\text{Loss}(y, \hat{y}) = (y - \hat{y})^2$$

the negative gradient is equal to (two times) the **residual**

$$-\nabla_{\hat{y}} \text{Loss}(y, \hat{y}) = 2(y - \hat{y})$$



adding a sub-model to the ensemble

1. first find the γ_m that minimizes

$$\sum_{i=1}^{n} \text{Loss}\left[y_i, F_{m-1}(\boldsymbol{x}_i) + \boldsymbol{\gamma_m} h_m(\boldsymbol{x}_i)\right]$$

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2. γ_m is multiplied by a **learning rate** η to mitigate overfitting

main hyperparameters

sklearn.ensemble.GradientBoostingClassifier

class `sklearn.ensemble.` **`GradientBoostingClassifier`**(**, loss='deviance', learning_rate=0.1, n_estimators=100, subsample=1.0, criterion='friedman_mse', min_samples_split=2, min_samples_leaf=1, min_weight_fraction_leaf=0.0, max_depth=3, min_impurity_decrease=0.0, min_impurity_split=None, init=None, random_state=None, max_features=None, verbose=0, max_leaf_nodes=None, warm_start=False, validation_fraction=0.1, n_iter_no_change=None, tol=0.0001, ccp_alpha=0.0*) [source]

Gradient Boosting for classification.

GB builds an additive model in a forward stage-wise fashion; it allows for the optimization of arbitrary differentiable loss functions. In each stage `n_classes_` regression trees are fit on the negative gradient of the binomial or multinomial deviance loss function. Binary classification is a special case where only a single regression tree is induced.

Read more in the User Guide.

- **ensemble size** (`n_estimators`)
- **learning rate**
- as well as the usual decision tree hyperparameters. . .

gradient boosting implementations

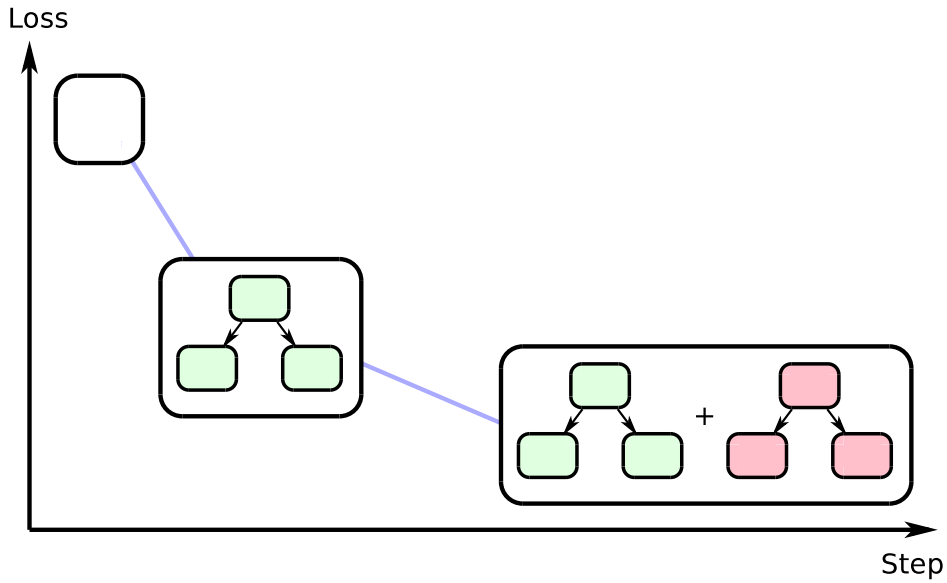
- scikit-learn (Python):
  - `sklearn.ensemble.GradientBoostingClassifier`
  - `sklearn.ensemble.GradientBoostingRegressor`
- **XGBoost** (multiple languages)
- **H2O** (multiple languages)
- ...

tree ensembles in Kaggle

- GB (and other tree-based ensembles) often dominate in competitions for various prediction tasks, such as Kaggle
- why?
  - robust to preprocessing
  - easy to mix different types of features, nice for tabular data
  - easy to tune
- especially the software XGBoost is popular in Kaggle

boosting: main points

- **boosting** algorithms build ensembles sequentially
- main idea: **correcting errors** from previous steps



- **gradient boosting** adds submodels based on a principle of gradient descent
- popular in industry and in competitions, mainly for tabular data